

Use Quarto for HW

↳ .qmd file

→ \$\$
math
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Actual Metrics

Hansen - Appendix A

↳ Vector a is $k \times 1$ list of numbers

$$\rightarrow a = \begin{pmatrix} a_1 \\ \vdots \\ a_k \end{pmatrix}$$

→ $k=1$; a is a scalar;

↳ Matrix A is $k \times r$

$$\rightarrow A = \begin{bmatrix} a_{11} & \cdots & a_{1r} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kr} \end{bmatrix}$$

→ A^T is $r \times k$

↳ square if $r=k$

→ symmetric if $A = A^T$

→ diagonal if all elements off diagonal = 0

↳ Identity matrix if all elements on diagonal equal 1

$$\rightarrow \underline{I}_k$$

↳ Matrix Addition:

$\rightarrow \exists \underline{A} = (a_{ij}), \underline{B} = (b_{ij})$ with same dimension, $\underline{A} + \underline{B} = (a_{ij} + b_{ij})$

$\rightarrow$ commutative: $\underline{A} + \underline{B} = \underline{B} + \underline{A}$

$\rightarrow$ associative: $(\underline{A} + \underline{B}) + \underline{C} = \underline{A} + (\underline{B} + \underline{C})$

↳ Matrix Multiplication:

$\rightarrow C$ is a scalar

$$\underline{A} \cdot C = C \cdot \underline{A} = (C \cdot a_{ij})$$

$\rightarrow$ if a and b are $k \times 1$ vectors:

↳ inner product $= a^T \cdot b = a_1 b_1 + \dots + a_k b_k$

$$= \sum_{j=1}^k a_j b_j$$

$$\rightarrow a^T b = b^T a$$

$\rightarrow a$ and b are orthogonal if $a^T \cdot b = 0$

↳ let $\underline{A}_{k \times k}$ $\underline{B}_{k \times s}$: conformable (can mult)

$$\rightarrow \underline{A} \cdot \underline{B} = \underbrace{\begin{bmatrix} a_1^T \\ \vdots \\ a_k^T \end{bmatrix}}_{k \times s} \cdot [b_1 \dots b_s] = \begin{bmatrix} a_1^T b_1 & \dots & a_1^T b_s \\ \vdots & \ddots & \vdots \\ a_k^T b_1 & \dots & a_k^T b_s \end{bmatrix}$$

↳ Matrix Mult not commutative
→ $\underline{A}\underline{B} \neq \underline{B}\underline{A}$

↳ Associative $\checkmark$ $(\underline{A} \cdot \underline{B}) \cdot \underline{C} = \underline{A} \cdot (\underline{B} \cdot \underline{C})$

↳ distributive $\checkmark$ $\underline{A}(\underline{B} + \underline{C}) = \underline{A} \cdot \underline{B} + \underline{A} \cdot \underline{C}$

↳ Trace of a $k \times k$ square matrix:
→ sum diagonal elements

$$\text{↳ } \text{tr}(\underline{A}) = \sum_{i=1}^k a_{ii}$$

→ $\exists \underline{A}, \underline{B}$, c is scalar.

↳ properties of trace

$$\rightarrow \text{tr}(c\underline{A}) = c \text{tr}(\underline{A})$$

$$\rightarrow \text{tr}(\underline{A}^T) = \text{tr}(\underline{A})$$

$$\rightarrow \text{tr}(\underline{A} + \underline{B}) = \text{tr}(\underline{A}) + \text{tr}(\underline{B})$$

$$\rightarrow \text{tr}(\underline{I}_k) = k$$

$$\rightarrow \underline{A}_{k \times r}; \underline{B}_{r \times k} : \text{tr}(\underline{A} \cdot \underline{B}) = \text{tr}(\underline{B} \cdot \underline{A})$$

$$\begin{aligned} \rightarrow \text{tr}(\underline{A} \cdot \underline{B}) &= \begin{bmatrix} a_1^T b_1 & \dots & a_1^T b_k \\ \vdots & \ddots & \vdots \\ a_k^T b_1 & \dots & a_k^T b_k \end{bmatrix} = \sum_{i=1}^k a_i^T b_i \\ &= \sum_{i=1}^k b_i^T a_i \\ &= \text{tr}(\underline{B} \cdot \underline{A}) \end{aligned}$$

↳ rank of $\underline{A}_{k \times r}$ s.t. $r \leq k$ is number of lin. indep col's. $\text{rank}(\underline{A})$. If $\text{rank}(\underline{A}) = r$, then $\underline{A}$ is full rank

→ linear independence means there is no non-zero $k \times 1$ vector $\underline{c}$ s.t.
 $\underline{A}^T \cdot \underline{c} = 0$

$$\text{rank} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}; \text{rank} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

" "
 1 2

$$\rightarrow \underline{c} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

↳ $\underline{A}_{k \times k}$ is non-singular if $\text{rank}(\underline{A}) = k$

↳ if $\underline{A}$ is non-singular $\exists \underline{A}^{-1}$

↳ $\underline{A}^{-1}$ satisfies $\underline{A} \cdot \underline{A}^{-1} = \underline{A}^{-1} \cdot \underline{A} = \underline{I}_k$

↳ for 2 non-singular matrices $\underline{A}, \underline{C}$

$$\rightarrow (\underline{A}^{-1})^T = (\underline{A}^T)^{-1}$$

$$\rightarrow (\underline{A} \cdot \underline{C})^{-1} = \underline{C}^{-1} \cdot \underline{A}^{-1} \quad \left. \vphantom{\rightarrow (\underline{A} \cdot \underline{C})^{-1} = \underline{C}^{-1} \cdot \underline{A}^{-1}} \right\} \text{square matrices}$$